

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 – 2019 (Odd Semester)

Innovativeness in Faculty Learning & Teaching and Learning outcome by Students

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC6502 Principles of Digital Signal Processing

Name of the Faculty member: Dr.S.Periyanayagi

Innovative Method	Topic	Effectiveness
Introduction	Introducing to the subject	It helps to avoid the fear on this subject.
Z to A Approach	Digital FIR and IIR Filter Design: Win amp with digital filter coefficients	It makes the concept clear and long lasting memory: While explaining this concept with designing of filter, it helps the students to understand the importance of filter coefficients
Application oriented teaching	Filter design, Convolution Adaptive Filters	• It helps to understand the importance of Filter designing in digital signal processing. • Adaptive filters in communication receivers and Medical fields. • Explained the concept of limit cycle oscillations with DSP processor using in laboratory make the students to understand the concept easily.
Analogy	Basic Filtering Concept	Basic filtering (new information) is explained with a concept of water filter and using mobile phones. (Known information).
Visualization PPT	Digital IIR filter design	It helped the students to understand the difference between analog and digital filter specification.
Simulation Tool (MATLAB)	Digital Filter design, Convolution, Interpolation, Decimation	Made the students to simulate various programs in the laboratory classes for the values of problems solved in the theory class. While simulating they experienced the concepts visually.

Giving Hints for solving problems	Digital IIR Filter for Butterworth and Chebyshev filters	Helps to recall the follow of problem easily in the Examination hall.
Known to unknown	(i) Discrete Fourier Transform and Fast Fourier Transform (ii) Analog Filters (iii) Quantization and Limit cycle oscillations	(i) The concept of DFT and FFT are explained from the concept of DTFT which they have already learnt in Signals and System (ii) The concept of filter designing is explained from Linear integrated circuits (iii) Explained Quantization with the concept of ADC (Known) which helps the students to understand the use of quantization. This creates the interest towards the subject.
Collaborative Learning	Solving problems in groups during tutorial hours. University Question and Answers for Unit I – Unit V	It helps the students to avoid careless mistakes in examination.

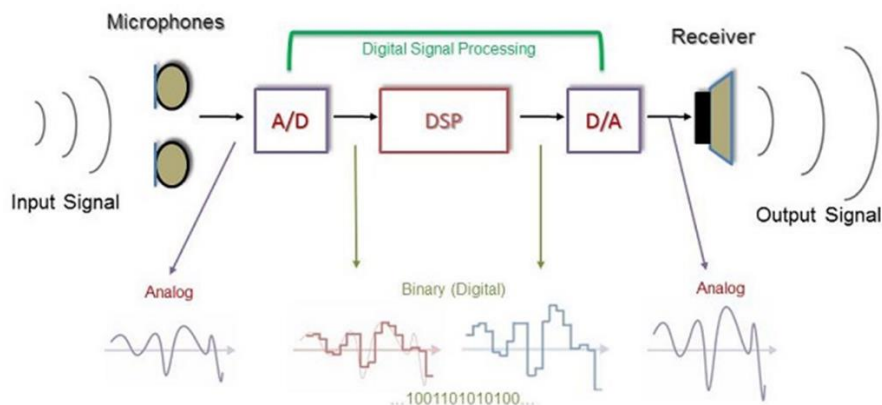
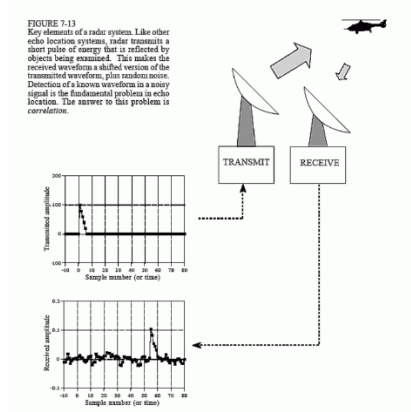
Using the concepts of online course in the subject:

Online Course: MITx.18.01.2x: Calculus 1B: Integration

Course Code and Title	Lecture No.	Title
EC6502 Principles of Digital Signal Processing	21	IIR Butterworth Filter Design by impulse invariant transformation.
	22	IIR Chebychev Filter Design by impulse invariant transformation.
	28	Structure of IIR – Cascade and Parallel form Realization.
	34&35	Filter design using windowing techniques- Rectangular/ Hanning Window
	36	Filter design using windowing techniques Hamming Window

Application of DSP:

- Used **Adobe Photoshop** to make sure you look good in your profile pic? Shake hands with Digital Image Processing, a brother of DSP



References:

1. John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
2. S.K. Mitra, 'Digital Signal Processing – A Computer Based Approach', McGrawHill Edu, 2013.
3. https://www.youtube.com/watch?v=HJ_-5mqUZ70
5. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-341-discrete-time-signal-processing-fall-2005/download-course-materials>